KLE Technological University

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Dr. M. S. Sheshgiri Campus, Belagavi

Department of
Electronics and Communication Engineering

MINOR Project Report

on

Automatic Seed Sowing and Spraying Agriculture Robot

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that project entitled **“Automatic Seed Sowing and Spraying Agriculture Robot”** is a bonafide work carried out by the student team of **“Chandrashekhar Bhavi-02FE22BEC020, Jamuna Pulpatli-02FE22BEC032, Priyanka Budihal-02FE22BEC054,Pruthviraj Banoshi-02FE22BEC058”**.The project report has been approved as it satisfies the requirements with respect to the minor project work prescribed by the university curriculum for B.E. (VI Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2025-2026.

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ABSTRACT

Modern agriculture demands increased efficiency, precision, and reduced reliance on manual labor to meet the growing food requirements of the global population. This project presents the design and development of an Automatic Seed Sowing and Spraying Agriculture Robot, aimed at automating two essential farming operations: sowing seeds and spraying pesticides. The robot is engineered to traverse agricultural fields with minimal human intervention, accurately placing seeds at predefined intervals and spraying pesticides evenly across the crop area.The system integrates mechanical movement with embedded electronics and control units to perform tasks with consistency and precision. By automating these processes, the robot not only reduces labor costs but also enhances productivity and supports sustainable agricultural practices. The report details the system architecture, hardware and software implementation, field testing, and the overall performance evaluation of the robot..

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Chapter 1

Introduction

1.1 Problem Statement

Design an autonomous robot capable of efficiently performing seed sowing tasks, including digging soil, sowing seeds, closing the soil, and spraying water. The robot must operate autonomously, ensuring precise seed placement, accurate soil coverage, and effective watering. It should adapt to various soil types and environmental conditions, while being energy-efficient for long-duration use. The system should integrate mechanical, electrical, and software components to provide reliable, high-precision performance in agricultural fields.

1.2 Introduction

The agriculture sector is the backbone of many economies around the world, yet it continues to face significant challenges such as labor shortages, low efficiency, and inconsistent crop productivity. With the advancement of technology, automation in agriculture has emerged as a viable solution to address these issues. This project, titled Automatic Seed Sowing and Spraying Agriculture Robot, aims to design and develop a multifunctional robotic system that automates two of the most labor-intensive tasks in farming: seed sowing and pesticide spraying.

The primary objective of this project is to enhance agricultural productivity by minimizing manual intervention, ensuring precise seed placement. The robot integrates mechanical systems with embedded electronics and control units to navigate agricultural fields autonomously or semi-autonomously, perform uniform sowing, and spray water efficiently. By reducing dependency on manual labor and ensuring precision in operations, this project contributes to sustainable farming practices and supports the growing need for smart agriculture.

This report outlines the design methodology, component selection, system architecture, fabrication process, and testing results of the prototype. The project reflects a significant step toward the integration of robotics in conventional farming and serves as a foundation for further development in automated agricultural solutions.

1.3 Objectives

The primary objective of the project is to design and develop an efficient, automated seed-sowing machine that caters to the agricultural needs of small and medium-scale farmers, particularly in resource-constrained settings. This machine aims to:

1.Reduce Human Labor: Traditional seed sowing involves labor-intensive tasks such as manual plowing, seed distribution, and covering, which require significant time and physical effort.

2.Improve Productivity: The machine ensures precise placement of seeds, maintaining uniform spacing and consistent depth, which are critical for optimal germination and growth.

3.Enhance Affordability: Most modern farming equipment is prohibitively expensive for small-scale farmers, limiting their access to advanced technologies. This project prioritizes cost-effective design and materials, making the machine a viable option for farmers with limited financial resources.

4.Increase Efficiency: The machine integrates multiple farming operations, such as soil preparation, seed planting, and row spacing, into a single automated system.

5.Enable Customization: Different crops and soil types have specific planting requirements, such as varying seed sizes, spacing, and depth.

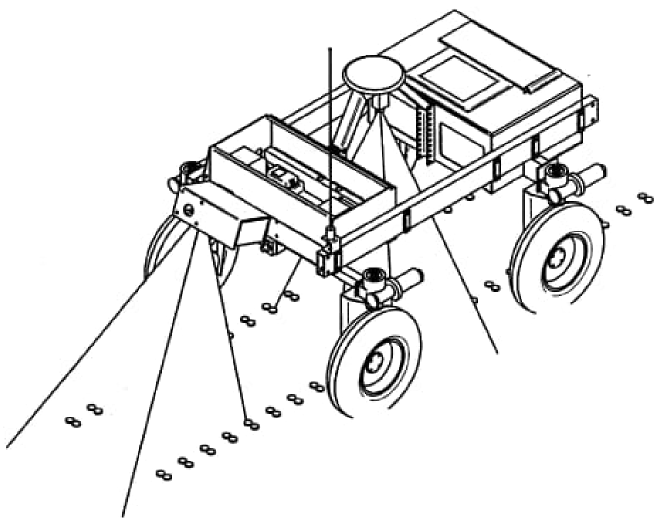

Figure 1.1: picture

Chapter 2

Literature Survey

2.1 Automatic Seed Sowing Robot

The Automatic Seed Sowing Robot is designed to automate the traditional seed sowing process in agriculture, which is typically manual and labour-intensive. This robot uses four motors for precise movement across a field, with each motor controlling the robot's direction and speed. It is equipped with a seed distribution mechanism consisting of a funnel that collects seeds and a shaft with gear-like teeth that efficiently dispenses the seeds into the soil. The system also features a bent plate to create a groove or slot in the ground for the seeds, ensuring they are placed at the correct depth. A tail-like rod follows, which covers the seeds with soil after sowing, completing the process. The robot is designed for small to medium-scale agricultural applications and aims to make farming more efficient by reducing the time and labor required for planting seeds. The benefits of the Automatic Seed Sowing Robot are significant, especially in terms of increasing the efficiency of farming processes and reducing human labour costs. The robot is capable of working in a consistent, controlled manner, ensuring uniform seed distribution across a field. This can result in more even crop growth and higher yields. The design and mechanism aim to reduce the risk of human error and soil damage, offering a more precise alternative to traditional seed sowing techniques. Additionally, the robot can operate for long hours, enabling continuous sowing without requiring rest, further optimizing agricultural productivity.

Research Gap
The Automatic Seed Sowing Robot, designed to automate the seed sowing process, presents several research gaps that need to be addressed for enhanced functionality and wider applicability. One significant gap lies in seed placement precision. While the robot improves efficiency, it may not consistently ensure precise seed placement at the right depth and spacing, which are crucial factors for optimal crop growth. Different seeds require specific planting conditions, and the robot's current system may not be adaptable enough to handle this variation. To enhance its performance, future research could focus on developing a more adaptable seed placement mechanism that can adjust to various seed types and environmental conditions. Another area for improvement is the robot's ability to operate across diverse soil types and terrains. The existing system may struggle with varying soil conditions, such as compacted, rocky, or uneven terrain, which could hinder its ability to perform efficiently across different fields. Enhancing the robot's design to

handle diverse soil types, as well as uneven or sloped terrains, would make it more versatile and capable of working in a broader range of agricultural environments. Research into improving its mobility system, such as wheels or treads, could help the robot better navigate these challenges. Energy efficiency is also a critical research gap, as the robot's reliance on electricity or battery power may limit its use in large-scale farming operations. For extended or large-field applications, energy sustainability becomes a concern. Addressing this gap could involve integrating renewable energy solutions, such as solar power, to ensure that the robot can operate for longer periods without frequent recharging or environmental impact. Exploring alternative energy sources or hybrid systems could make the robot more energy-efficient and cost-effective in the long run.

2.2 Design and Fabrication of Manually Operated Seed Sowing Robot

The paper on the "Design and Fabrication of Manually Operated Seed Sowing Robot" presents a promising solution to reduce manual labor and improve seed sowing efficiency, but it also highlights several research gaps that need to be addressed for broader applicability. One major gap is the system's reliance on manual operation, which limits its potential for full automation and prevents it from achieving higher levels of precision and efficiency. The robot lacks advanced technologies such as sensors, autonomous navigation systems, or motorized controls, which could significantly enhance its accuracy in seed placement and depth. Additionally, the robot's design may not adapt well to uneven, sloped, or irregular terrains, restricting its use to fields with uniform and flat surfaces. This lack of adaptability reduces its versatility for farmers working in diverse agricultural landscapes. Another significant limitation is the robot's scalability. While the current design is suitable for small-scale farming, it may not be practical for large-scale agricultural operations where speed and capacity are crucial. Moreover, the absence of energy-efficient solutions, such as solar power integration, reduces its sustainability, especially for prolonged usage. The system also lacks real-time monitoring capabilities, such as feedback mechanisms to assess seed placement accuracy, soil conditions, or progress during field operations, which are critical for precision agriculture. Furthermore, the robot focuses solely on seed sowing and does not integrate additional functionalities like watering, pesticide spraying, or soil health monitoring, which would provide a more comprehensive agricultural solution. Addressing these gaps through automation, multi-tasking capabilities, and advanced technologies can make the robot more efficient, adaptable, and scalable for modern farming needs.

Research Gap

The paper focuses on the development of a manually operated seed sowing robot aimed at improving the efficiency of traditional farming practices. The robot is designed to simplify and automate seed sowing tasks, ensuring consistent seed placement, reducing labour intensity, and enhancing productivity. The design incorporates a lightweight frame, wheels for mobility, and a seed metering mechanism for uniform seed distribution. The robot is operated manually, eliminating the need for complex electronics or power systems, which makes it more accessible to small-scale farmers. This system reduces seed wastage and labour costs while maintaining accuracy and ease of use.

2.3 Agribot: Arduino Controlled Automatic Seed Sowing Machine for Small Scale Cultivation

Avinash kumar Parmar, Talati Chaitanya J.(2024) The paper "Agribot: Arduino Controlled Automatic Seed Sowing Machine for Small Scale Cultivation" focuses on addressing the challenges faced by small-scale farmers during the sowing process through the development of a cost-effective, automated solution. Traditional methods of manual sowing are labour-intensive, time-consuming, and prone to inefficiencies like uneven seed placement and suboptimal germination rates, which impact overall crop yields. The study proposes the "Agribot," an Arduino-controlled robot designed to automate multiple stages of cultivation, including soil preparation, seed placement, and water application. The Agribot integrates several features to enhance productivity and reduce manual labour. The system is powered by a battery and controlled using an Arduino microcontroller, ensuring cost-efficiency and ease of use. It employs DC motors to operate the seed rotor for precise seed dispensing and servo motors to direct water sprinklers, ensuring uniform seed placement and irrigation. Additionally, the machine is equipped with a cultivator to prepare the soil, facilitating better seed germination conditions. The Agribot operates autonomously across the field, dispensing seeds at fixed intervals to ensure consistent spacing, while also performing watering functions. The design emphasizes affordability and simplicity, making it accessible to small-scale farmers, and aims to address inefficiencies associated with traditional methods.

Research Gap

The Agribot seeks to bridge these gaps by providing a low-cost, versatile, and efficient alternative tailored specifically for small-scale cultivation. Its ability to integrate multiple farming functions—soil preparation, seed sowing, and watering—into a single platform reduces the need for additional machinery, lowers operational costs, and saves time. By leveraging automation through an Arduino-based control system, the Agribot offers a scalable solution that is easy to operate, even for farmers with minimal technical expertise. The research contributes to advancing precision agriculture and promoting sustainable practices, particularly in resource-constrained environments.

2.4 Design And Fabrication of Automatic Seed Sowing Machine Using Iot

The paper titled "Design and Fabrication of Automatic Seed Sowing Machine Using IoT" focuses on creating an innovative, automated system to address inefficiencies in traditional seed sowing practices. The proposed machine integrates robotics with IoT technology to offer precise and consistent seed sowing, thus reducing labor intensity and time consumption. The system employs an Arduino microcontroller to execute programmed operations, with a Bluetooth module facilitating wireless control through a mobile application. Other key components include a Viper motor, chain and sprocket mechanism, pillow block bearings, and a funnel, which together ensure smooth operation and uniform seed dispersion. The machine functions by receiving commands from a mobile device, enabling the precise placement of seeds with uniform spacing and depth. Its IoT integration not only allows remote operation but also supports monitoring and data analysis for better management. The study demonstrates that this automation significantly improves sowing efficiency, achieving a sowing rate of 30–35 seeds per minute, a spacing of 10 cm, and an optimal depth of 1.5 cm. These features highlight the system's potential to transform traditional agricultural practices. The methodology adopted for the project includes defining clear objectives, conducting a thorough literature review, and conceptualizing the design. A prototype of the machine was developed and tested in different field conditions. Results show that the machine reduces manual labor, increases precision, and ensures uniform seed placement. However, the machine's performance is impacted by extreme soil moisture conditions, making it less effective on overly dry or wet soils.

Research Gap

The study identifies several gaps in current seed sowing technologies and seeks to address them. A major limitation of existing systems is their rigid seeding patterns, which often allow only a single type of seed to be sown. The designed machine overcomes this constraint with adaptable mechanisms, enabling broader functionality. Additionally, many traditional methods lack IoT integration, limiting their efficiency and adaptability. The proposed machine bridges this gap by incorporating IoT features that allow remote monitoring and control, significantly enhancing usability. Precision and uniformity in seed placement are persistent challenges in manual and semi automated sowing methods. The developed system resolves this issue by ensuring consistent spacing and depth during seed placement. However, the machine still struggles to perform efficiently under extreme soil conditions, especially when the soil is excessively wet or dry. This limitation points to the need for further research into soil adaptability through advanced sensing mechanisms. Future research could explore improving the machine's environmental adaptability by incorporating sensors to measure and respond to varying soil conditions. Additionally, expanding IoT functionalities to provide real-time data insights on soil health and crop yield could further enhance the system's utility. Lastly, efforts to reduce costs and improve scalability are essential for making the technology accessible to resource-limited farming communities.

2.5 Automatic Seed Sowing and Spraying Agriculture Robot

The methodology for developing an automatic seed sowing and spraying agricultural robot involves several key steps. Initially, the robot's dimensions are determined based on specific design constraints. Subsequently, components such as sensing devices, actuators, seed handling units, microprocessors, stepper motors, servomotors, and communication modules are selected and integrated. The design process utilizes CAD software like PTC CREO 3.0 to create detailed models of the robot. The robot's operation is managed via a Wi-Fi module, enabling remote control over large distances through IoT-enabled devices. This setup allows the robot to perform tasks such as ploughing, seed sowing, and field moistening autonomously, thereby reducing manual labor and enhancing agricultural efficiency. The methodology for the automatic seed sowing and spraying agriculture robot involves a systematic integration of hardware and software components to perform multiple tasks autonomously. The design begins with conceptualizing the robot's structure, taking into account the functional requirements and field conditions. A robust yet lightweight frame is constructed to house the components, ensuring portability and durability. The hardware includes a seed sowing mechanism driven by stepper motors and servos, enabling precise placement of seeds at desired intervals. A spraying unit equipped with a submersible pump and nozzle ensures uniform distribution of pesticides or fertilizers. Sensors for soil moisture, temperature, and humidity are integrated to gather real-time data about field conditions, which is processed by a microcontroller unit (e.g., Arduino or similar). The robot's operation is facilitated by IoT technology, allowing remote control via a smartphone or computer through a Wi-Fi module. The use of CAD software, such as PTC CREO 3.0, ensures accurate design and assembly of the robot's components. Software algorithms are developed to automate tasks like ploughing, seed dispensing, and spraying based on environmental conditions and predefined parameters. Solar panels or rechargeable batteries are used for powering the robot, ensuring energy efficiency and sustainability. This methodology enables the robot to perform tasks with minimal human intervention, improving precision, reducing labor costs, and promoting smart agricultural practices.

Research Gap

The research paper highlights significant advancements in the automation of agricultural tasks, such as seed sowing and pesticide spraying, but also reveals several research gaps that can guide future studies. One notable gap is the limited adaptability of the robot to diverse terrains and varying crop requirements. While the system is designed for specific operational conditions, it lacks the flexibility to handle irregular fields, steep slopes, or diverse soil types, which are common in real-world agricultural settings. Additionally, the robot's reliance on predefined parameters for seed placement and pesticide spraying may not account for dynamic environmental changes, such as sudden weather variations or pest infestations, which require real-time decision-making and adaptability. Another gap lies in the scalability of the system for larger fields or diverse crop types. The current design and sensor integration may not efficiently handle large-scale farming operations or cater to crops with unique planting and spraying needs. Energy efficiency, although addressed through solar panels and rechargeable batteries, could be further optimized to ensure uninterrupted operations over extended periods, especially in areas with limited sunlight. Furthermore, the communication system, relying on Wi-Fi modules, may face challenges in remote areas with poor connectivity, limiting the practical usability of IoT-enabled features. Finally, the paper does not extensively address cost-effectiveness, which is a critical factor for widespread adoption, particularly in resource-constrained farming communities. The integration of advanced technologies, such as machine learning for adaptive decision making or more robust communication protocols like LoRa or 5G, is also unexplored. These research gaps underscore the need for future developments to enhance the robot's adaptability, scalability, and affordability, ensuring its practicality and accessibility for a broader range of agricultural scenarios.

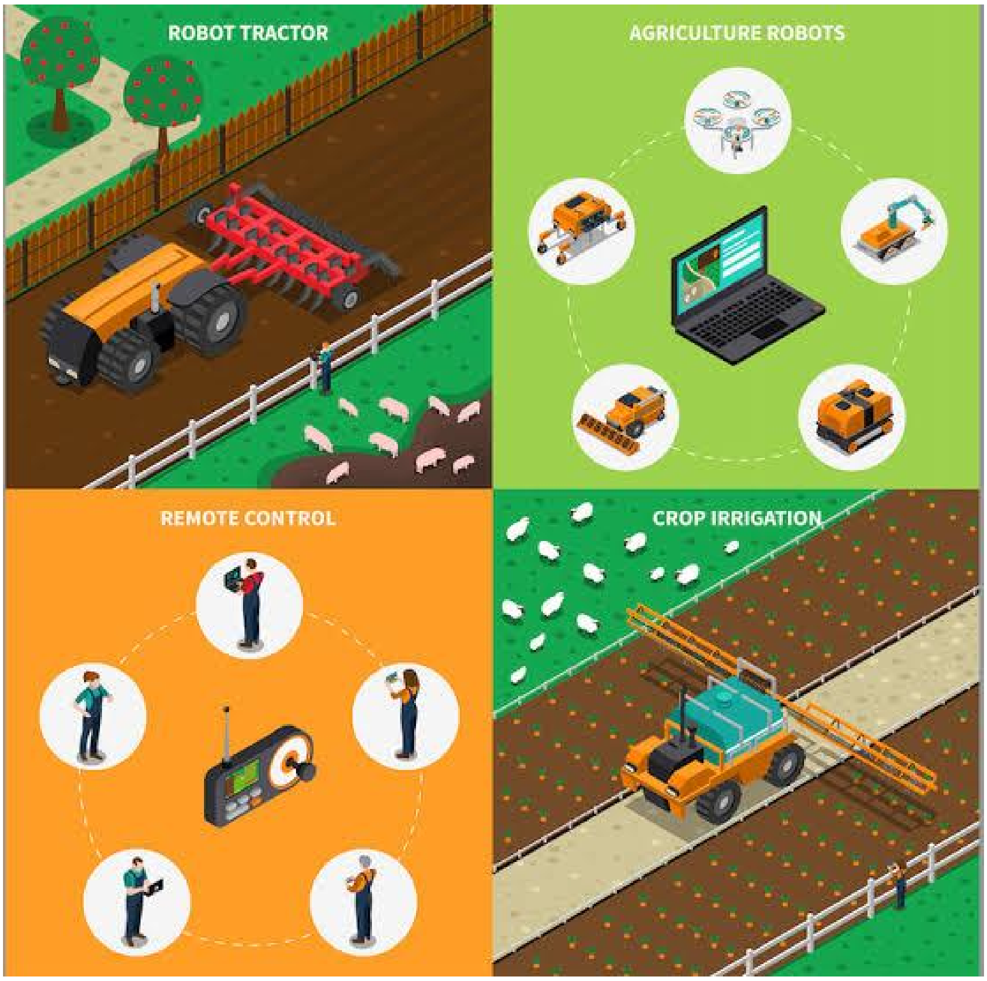

Figure 2.1: picture

Chapter 3

Project Planning

3.1 Gantt Chart

A Gantt chart is a visual project management tool that represents the timeline of a project. It displays tasks or activities as horizontal bars along a timeline, with the length of each bar indicating the duration of a task.The Gantt chart helps project managers track progress, manage deadlines, and visualize how tasks overlap or depend on one another. It is widely used for scheduling, resource allocation, and ensuring timely completion of projects.

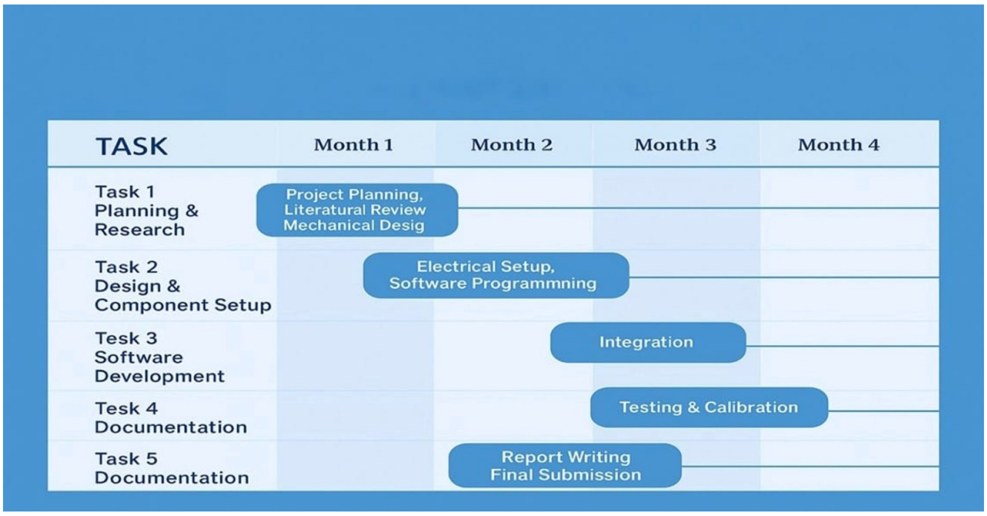

Figure 3.1: Gantt chart

3.2 Work Breakdown Structure

A Work Breakdown Structure (WBS) is a project management tool that divides a project into smaller, manageable components or tasks. It organizes the work into a hierarchy, making it easier to plan, assign resources, and track progress. The WBS helps clarify the scope of the project,ensuring that all required work is identified. It also enhances communication among team members and stakeholders, assigns responsibilities, and improves time and resource management.By breaking down complex projects into manageable parts, it makes it easier to estimate costs,schedule timelines, and monitor progress.

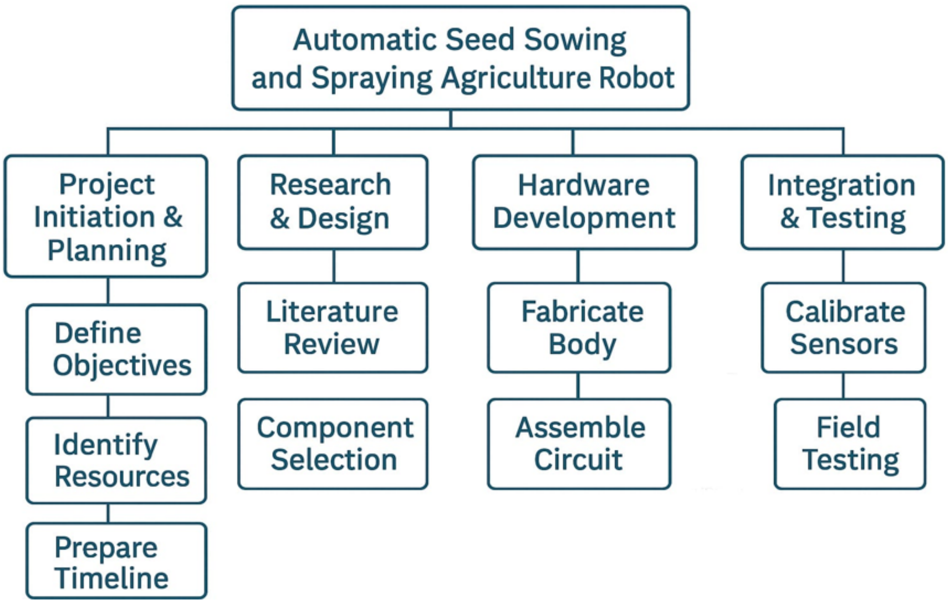

Figure 3.2: Work Breakdown Structure

Chapter 4

Design Specification

4.1 Specification

1.Microcontroller: Arduino Mega

2.Chassis: 2 or 4-wheel platform

3.Motors: DC/Servo Motors with L298N driver

4.Sensors: IR, Ultrasonic

5.Seed Mechanism: Servo-controlled dropper

6.Power Supply: 12V Li-ion

7.Battery Speed: 0.5 m/s (adjustable)

8.Seeding Interval: Configurable (e.g., every 10–20 cm)

9.Field Suitability: Small experimental plots

Chapter 5

Methodology

The design and implementation of the autonomous seed sowing machine involve theintegration of mechanical, electrical, and software components that work together to achieve the desired functionality: digging, sowing, covering, and watering. The robot is built with various subsystems, each serving a critical role in the execution of the tasks in an efficient and precise manner. The methodology is described in the following sections, covering the mechanical design, electrical and control systems, and software programming.

5.1 Detailed Block Diagram

A detailed Block Diagram is a graphical representation of the functions and operations within a system, showing how the various components interact to achieve a specific objective. It is widely used in engineering, systems design, and project planning to depict the relationships and flows between different parts of a system.

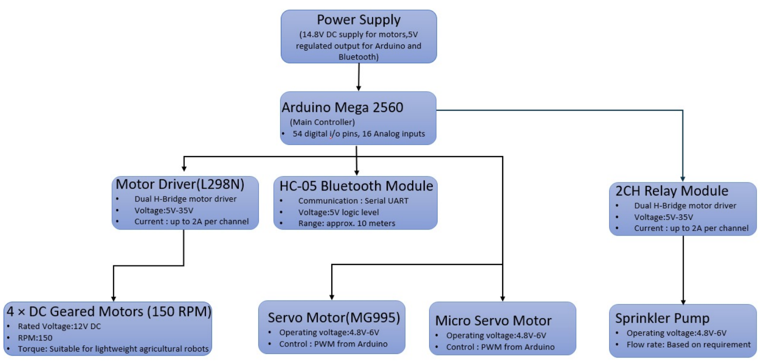

Figure 5.1: Block Diagram

5.2 Block Diagram Specification

1. **Power Supply**
   - Input: 14.8V DC
   - Output: 14.8V DC for motors, 5V regulated output for Arduino and Bluetooth
   - Purpose: Supplies power to all components in the system.
2. **Arduino Mega 2560 (Main Controller)**
   - Function: Central control unit
   - I/O Pins: 54 digital input/output pins 16 Analog input pins
   - Power Supply: 5V regulated from power module
   - Communication: Controls peripherals via digital signals and PWM
3. **4 $\times$ DC Geared Motors (150 RPM)**
   - Rated Voltage: 12V DC
   - Speed: 150 RPM
   - Use: Lightweight agricultural robot movement
   - Torque: Moderate, suitable for mobile platforms
4. **HC-05 Bluetooth Module**
   - Communication: Serial UART
   - Voltage Level: 5V logic level
   - Range: 10 meters
   - Purpose: Wireless communication with Arduino (e.g., from smartphone or computer)
5. **2CH Relay Module**
   - Type: 2-channel relay board
   - Voltage Support: 5V to 35V
   - Current Capacity: Up to 2A per channel
   - Use: Switching high-power devices like pump or servos
6. **Micro Servo Motor**
   - Operating Voltage: 4.8V to 6V
   - Control: PWM from Arduino
   - Use: Precise, lightweight motion control
7. **Sprinkler Pump**
   - Operating Voltage: 4.8V to 6V
   - Control: Likely through relay module

- Flow Rate: Configurable based on specific needs
- Use: Water distribution/spraying in agricultural setup

8. **Motor Driver (L298N)**

- Type: Dual H-Bridge motor driver
- Voltage Range: 5V to 35V
- Current: Up to 2A per channel
- Controlled Device: DC geared motors

9. **Servo Motor (MG995)**

- Operating Voltage: 4.8V to 6V
- Control: PWM from Arduino : Typically for steering or controlled positioning

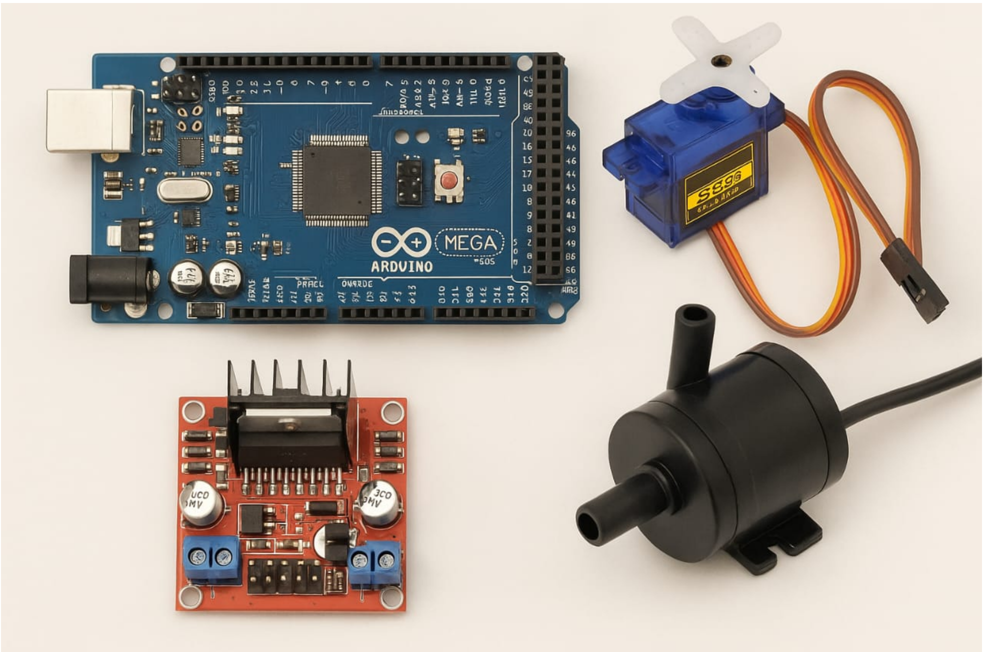

Figure 5.2: components

Chapter 6

Bill of Material

Sl.No	Material	Quantity	Price
1	Arduino Mega 2560	1	1,199
2	L298N Motor driver	1	200
3	2 Channel Relay	1	150
4	LM 25965 DC -DC 35V-35	1	150
5	Power supply 14.8v (4 * 3.7V)	4	240
6	2 x 3.7V AAA Battery cell holder	2	60
7	4 DC motor 150 RPM	4	1,000
8	Micro servo motor (SG90)	1	200
9	Metal Gear servo motor (MG90S)	1	200
10	5V Water Pump	1	140
11	Water Pump pipe	1	60
12	PVC pipe	1	100
13	Led white	2	10
14	Led red	2	10
15	270k ohm Resisters	4	10
16	DC Motor Holder	4	100
17	Chase	1	150
18	Jumper wires(male-male, female-male ,female-female)	few	200
19	Single standard wire	few	40
	Total		**4,219**

Table 6.1: Itemized List of Materials and Costs

Chapter 7

Results

The autonomous seed sowing robot achieved its intended goals, demonstrating significant improvements in efficiency, precision, and resource optimization for agricultural tasks. The system's ability to perform multiple functions digging, sowing, covering, and watering seamlessly and efficiently highlights its potential to revolutionize traditional farming practices. While certain limitations were observed during testing, these are addressable with future refinements, such as adaptive mechanisms, better calibration for varying seed sizes, and extended communication capabilities. Overall, the results confirm the robot's viability as a valuable tool for modern, automated agriculture.

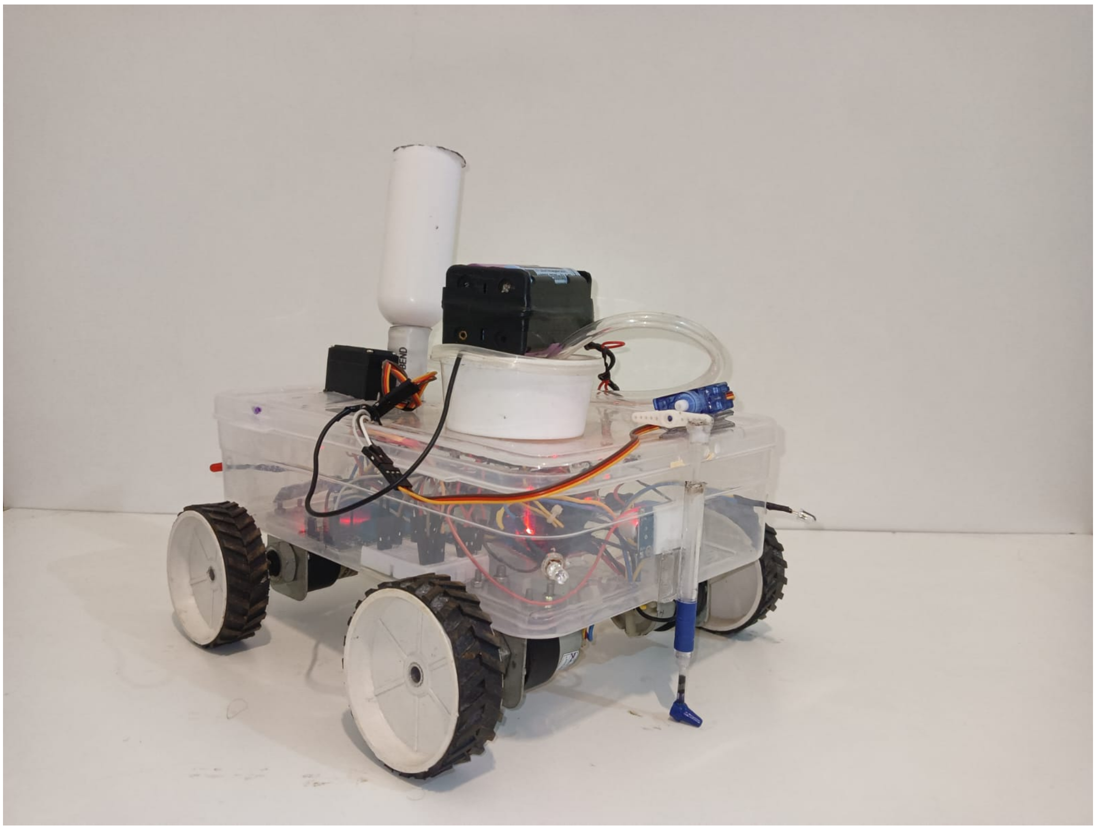

Figure 7.1: Result

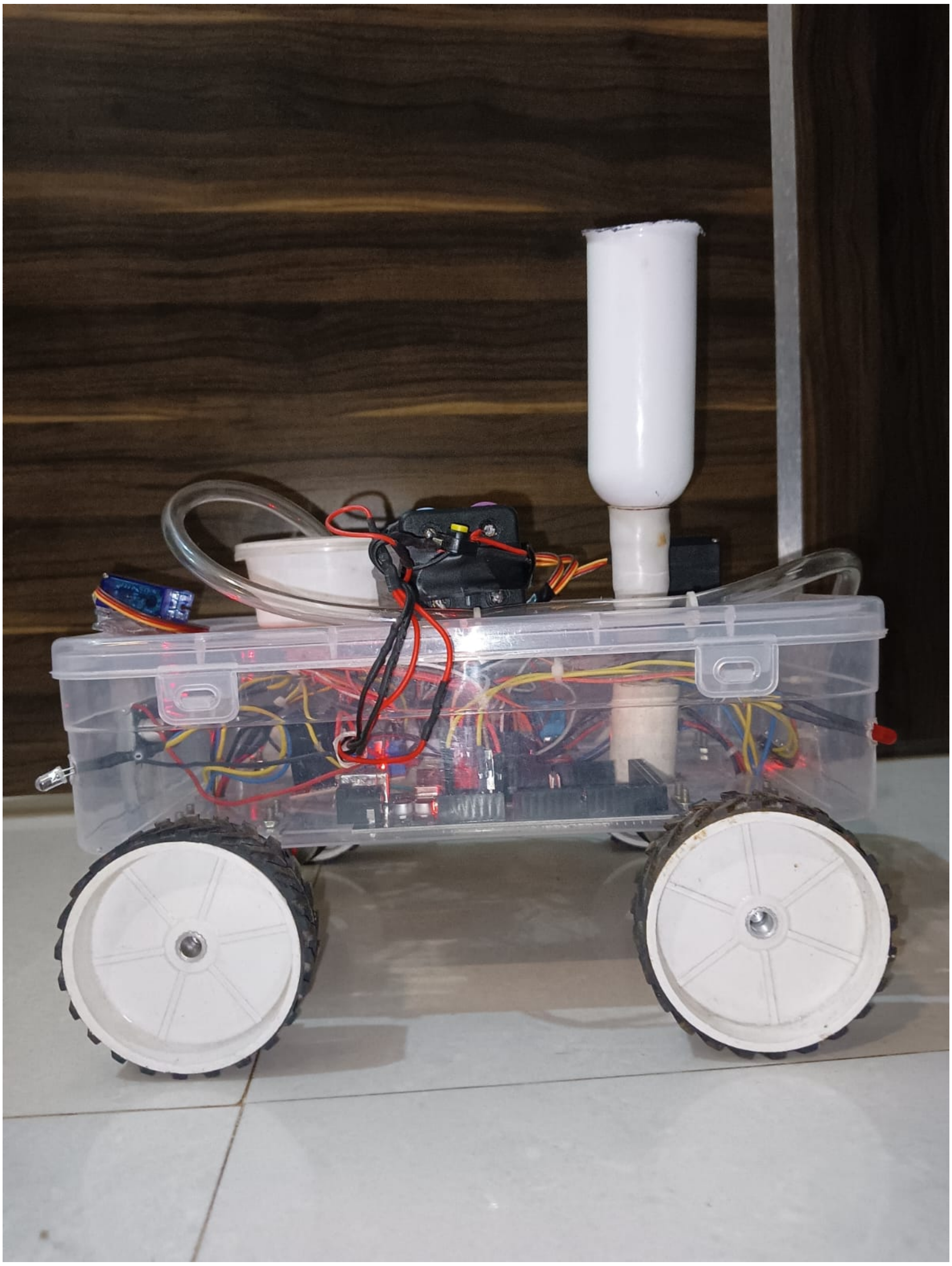

Figure 7.2: Result

Chapter 8

Conclusion and Future scope

8.1 Conclusion

The autonomous seed sowing robot marks a significant advancement in agricultural automation by integrating soil digging, seed sowing, soil covering, and water spraying into a single, efficient system. The robot demonstrated reliable performance during testing, with precise seed placement, consistent soil coverage, and effective water distribution, significantly reducing manual effort and improving productivity. Its user-friendly Bluetoothbased control system allowed real-time adjustments for seed spacing and water flow, catering to various crop requirements. Despite challenges such as limited adaptability to uneven terrains, handling seeds of varying sizes, and the communication range of the control system, the project highlights a promising solution to modern agricultural challenges like labor shortages and inefficiencies. With further refinements, including terrain adaptability, extended-range connectivity, and advanced navigation systems, the robot has the potential to revolutionize farming practices, contributing to higher yields, resource optimization, and sustainable agricultural development.

8.2 Future Scope

The autonomous seed sowing robot provides a strong foundation for integrating automation in agriculture, but several enhancements can be explored to expand its functionality and improve its performance. One major area for improvement is terrain adaptability. Developing a self-adjusting digging and soil-covering mechanism would enable the robot to operate efficiently on uneven or rocky terrains, ensuring consistent performance in diverse field conditions. Additionally, incorporating advanced navigation systems such as GPS or LiDAR would allow for precise path planning and autonomous movement, making the robot suitable for larger fields and reducing reliance on manual control. To overcome the communication range limitations of the Bluetooth control system, transitioning to long-range technologies such as LoRa or Wi-Fi could enable seamless operation in larger fields. Additionally, solar panels could be integrated to enhance energy independence, especially in remote or off-grid farming areas, extending the robot's operational time and sustainability. Overall, the future scope of this project lies in improving its precision, adaptability, and sustainability while expanding its functionalities.

Chapter 9

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